

Foundation (Jalapeno)

Q1. Solve the system: $x + y = 4$, $x^2 + y^2 = 16$

- (A) (0, 4)
 (B) (4, 0)
 (C) (2, 2)
 (D) (0, 0)
 (E) No solution

Q2. Find the intersection point of $y = 2x - 1$ and $y = x^2 - 4x + 3$

- (A) (1, 1)
 (B) (2, 3)
 (C) (3, 5)
 (D) (0, -1)
 (E) (4, 7)

Q3. A company's profit is given by $P = 2x - 5$ and cost is $C = x^2 + 2x + 1$. When is the profit equal to the cost?

- (A) $x = 1$
 (B) $x = 2$
 (C) $x = 3$
 (D) $x = 4$
 (E) $x = 5$

Q4. Solve for x and y : $x - y = 2$, $x^2 + y^2 = 20$

- (A) (4, 2)
 (B) (2, 4)
 (C) (3, 1)
 (D) (-2, -4)
 (E) (-4, -2)

Q5. Find the value of y when $x = 2$: $y = x^2 - 3x + 1$, $y = 2x - 5$

- (A) -1
 (B) 0
 (C) 1
 (D) 2
 (E) 3

Q6. What is the intersection point of $y = x + 2$ and $y = x^2 - 2x - 1$?

- (A) (1, 3)
 (B) (-1, 1)
 (C) (3, 5)
 (D) (0, 2)
 (E) (2, 4)

Q7. A stock's price is modeled by $P = 2x + 5$ and its volume by $V = x^2 + 3x + 2$. Find the point where $P = V$

- (A) (0, 2)
 (B) (1, 7)
 (C) (2, 12)
 (D) (-1, 0)
 (E) (3, 20)

Q8. Solve the system: $y = x^2 - 2x - 3$, $y = -x + 1$

- (A) (1, 0)
 (B) (-1, 2)
 (C) (3, 4)
 (D) (-2, 3)
 (E) (0, -3)

Open Ended Questions (FRQ)

Q9. A business's revenue is given by $R = 3x + 2$ and its cost by $C = x^2 + x + 1$. Find the point where $R = C$ and explain its significance. Provide calculations and show work.**Q10.** The price of a stock is modeled by $P = x^2 - 4x + 5$ and its volume by $V = 2x - 3$. Use the system of equations to find when the price equals the volume, and describe the practical implications. Show all work and provide a final answer.

Chef's Tips

- Emphasize the importance of checking solutions in the original equations.
- Encourage students to graph the equations to visualize the intersection points.
- Remind students that not all systems have real solutions.